SIMATS ENGINEERING ASSIGNMENT TOOL-5

COUSE NAME: Fundamentals of Data Science COURSE CODE: DSA0402

COURSE OUTCOME COVERED

CO5: Design and implement end-to-end data-driven solutions by integrating pre-processing, statistical analysis, and machine learning for real-world applications ***(BL3)***

Assessment Tool Weightage: 40%

QUESTIONS: 01 **Total Marks:100**

Annexure A
HACKATHON

Code of Conduct:

*I **Jannathul Firdaus. N (192573025)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Jannathul Firdaus. N

SCENARIO BASED ASSESSMENT							
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	Level	Marks
Understanding of Problem	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario		
Data Preparation and EDA	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues		
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts		
Decision Making	20%	Makes wellreasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided		
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand		

Annexure A

1. AI-Based Crop Advisory Reliability Assessment

Theme: Smart Agriculture
Farmers receive crop recommendations based on soil type, rainfall, temperature, humidity, crop history, fertilizer use, and pest incidents. However, one recommendation model may work well in irrigated regions but fail in drought-prone areas.
Hackathon Task:
Develop and evaluate multiple crop-advisory models for crop selection, fertilizer recommendation, or disease-risk prediction. Assess whether the selected model remains reliable across districts with different climatic and soil conditions.
Evaluation Focus:

- Compare Accuracy, F1-score, and region-wise performance.
- Assess whether recommendations are biased toward high-resource farming areas.
- Justify which model should be deployed for farmers and under which conditions human agricultural officers must review the output.

2. Smart Public Health Risk Prioritization

Theme: Healthcare and Biomedical Devices
A district health department collects data on age, symptoms, vaccination status, chronic diseases, hospital availability, travel history, and previous disease outbreaks. It wants to identify high-risk areas for disease spread and allocate medical resources efficiently.
Hackathon Task:
Build and evaluate predictive models for disease-risk prioritization at ward or village level. Recommend a decision-support strategy for allocating ambulances, medicines, and medical staff.
Evaluation Focus:

- Compare model performance for detecting high-risk regions.
- Evaluate the impact of false negatives, where a high-risk area is not identified.
- Assess fairness so that rural and low-data regions are not ignored.
- Justify the final resource-allocation policy.

3. Urban Traffic Congestion and Emergency Route Evaluation

Theme: Smart Mobility
A city collects GPS traffic data, road speed, accident reports, weather conditions, signal timings, public-event schedules, and ambulance movement records. The current traffic prediction system reduces travel time in normal hours but performs poorly during festivals, rain, and road closures.

Hackathon Task:
Develop and evaluate traffic prediction and emergency-route recommendation models. Recommend the most dependable model for real-time ambulance and fire-service routing.
Evaluation Focus:

- Compare prediction error during normal, peak, and emergency conditions.
- Assess travel-time reduction achieved by each routing approach.
- Evaluate the consequences of wrong route recommendations.
- Justify the most reliable deployment strategy.

4. Smart Waste Collection Prioritization System

Theme: Clean and Green Technology
A municipality uses smart-bin sensors, population density, waste type, collection history, weather information, and festival schedules to plan waste collection. Fixed collection schedules often lead to overflowing bins in crowded areas and unnecessary trips to empty bins.
Hackathon Task:
Create and evaluate a waste-bin overflow prediction model and route-prioritization system for collection vehicles.
Evaluation Focus:

- Compare baseline fixed schedules with data-driven schedules.
- Assess fuel consumption, travel distance, missed-bin rate, and overflow reduction.
- Evaluate whether low-income or remote areas receive fair service coverage.
- Recommend the most cost-effective and socially equitable solution.

5. Air Quality Prediction and Pollution-Control Decision Support

Theme: Environment and Sustainability

A city monitors PM2.5, PM10, CO , temperature, humidity, traffic density, industrial activity, construction work, wind speed, and rainfall. Authorities need early warnings before pollution levels become dangerous.
Hackathon Task:
Develop and evaluate air-quality forecasting models that predict pollution levels for the next 24 hours. Recommend practical interventions such as traffic restrictions, construction control, or public health alerts.
Evaluation Focus:

- Compare MAE, RMSE, and high-pollution-event detection rate.

- Assess the ability to predict severe pollution days rather than only average conditions.
- Evaluate the cost and social impact of proposed interventions.
- Justify the recommended response plan.

6. Smart Water Supply and Leakage Risk Assessment

Theme: Water Resources and Infrastructure
A municipal water board has data on pipeline age, pressure readings, flow rate, consumption pattern, repair history, soil type, rainfall, and reported leakage incidents. Water losses are increasing, especially in older residential zones.
Hackathon Task:
Develop and evaluate models that identify leakage-prone areas and prioritize pipeline inspection or replacement.
Evaluation Focus:

- Compare leakage prediction models using Precision, Recall, and F1-score.
- Assess the cost of unnecessary inspections versus undetected leaks.
- Evaluate whether the model performs equally well in old and newly developed areas.
- Recommend a maintenance-priority policy.

7. Learning Outcome and Dropout-Risk Evaluation for Government Schools

Theme: Smart Education
A school education department maintains records of attendance, examination marks, household income category, distance to school, digital-access availability, teacher-student ratio, scholarship support, and previous dropout records.
Hackathon Task:
Develop and evaluate student-risk prediction models that help schools identify learners who may need academic or financial support.
Evaluation Focus:

- Compare models for identifying students at dropout risk.
- Evaluate false-positive and false-negative implications.
- Assess fairness across gender, rural/urban region, and socioeconomic category.
- Recommend ethical intervention strategies without stigmatizing students.

8. Smart Disaster Early-Warning Reliability System

Theme: Disaster Management
A coastal district receives data from rainfall sensors, river-water levels, cyclone alerts, satellite images, soil moisture, dam discharge, population density, evacuation-centre capacity, and historical disaster records.
Hackathon Task:
Develop and evaluate models that predict flood, cyclone, or landslide risk and recommend evacuation priorities.
Evaluation Focus:

- Compare model performance for early identification of high-risk zones.
- Evaluate warning lead time and missed-event rate.
- Assess whether vulnerable communities receive adequate priority.
- Justify the final alert threshold and evacuation-support plan.

9. Public Distribution System Anomaly and Leakage Detection

Theme: Governance and Digital India
The Public Distribution System maintains data on ration-card transactions, beneficiary count, commodity allocation, shop inventory, distribution date, biometric verification, and historical complaints. Authorities suspect irregularities such as duplicate claims, abnormal stock movement, and delayed distribution.
Hackathon Task:
Develop and evaluate anomaly-detection methods to identify possible leakage or irregular distribution patterns.
Evaluation Focus:

- Compare rule-based, clustering-based, and machine-learning approaches.
- Assess false-alarm rates because incorrect identification can affect fair-price shop owners.
- Evaluate transparency and explainability of the flagged cases.
- Recommend an audit process that combines analytics with human verification.

10. Smart Energy Demand Forecasting for Public Buildings

Theme: Renewable Energy and Smart Infrastructure
Government offices, schools, hospitals, and public transport facilities consume electricity based on occupancy, weather, working hours, equipment usage, solar generation, and seasonal conditions. Energy wastage is high because consumption is not actively monitored.
Hackathon Task:
Develop and evaluate energy-demand forecasting models and recommend energy-saving actions for public buildings.

Evaluation Focus:

- Compare models using MAE, RMSE, and peak-demand prediction accuracy.
- Assess electricity-cost savings from each proposed action.
- Evaluate whether solar energy and battery usage improve reliability.
- Justify the final energy-management recommendation.

Question 2: Smart Public Health Risk Prioritization

1. Understanding of the Problem

A district health department wants to know which wards or villages are most likely to become high-risk areas for disease spread, so that ambulances, medicines and medical staff can be sent where they will prevent the most harm. The model is a decision-support tool for officers, not a replacement for them.

- **Machine-learning task.** Binary classification at ward/village level: high-risk (1) or not high-risk (0). The predicted probability is also used to rank wards for resource allocation.
- **Key issues.** (i) High-risk wards are a minority, so accuracy alone is misleading; (ii) a missed high-risk ward (false negative) costs far more than an unnecessary alert (false positive); (iii) rural wards have missing and under-reported data, so a model can quietly ignore them; (iv) the final policy must be explainable to officers.
- **Success criteria.** Recall of at least 90% for high-risk wards overall and separately for rural and urban wards, acceptable precision, and a clear allocation rule.

2. Data Preparation and Exploratory Analysis

A simulated ward-level dataset of 1,200 wards was created (540 rural, 45.0%). Each ward has seven predictors and a rural/urban flag; the target is whether the ward was classed as high-risk.

Variable	Meaning
pct_elderly	Share of population aged 60+ (%)
symptom_rate	Reported fever/respiratory symptoms per 1,000 residents
vaccination_cov	Vaccination coverage (%)
chronic_rate	Residents with chronic disease (%)
beds_per_1000	Hospital beds available per 1,000 residents
travel_inflow	Inbound travellers per day (hundreds), a measure of travel history exposure
past_outbreaks	Outbreaks recorded in the ward in earlier years
rural	1 = rural/low-data ward (used for fairness monitoring)
high_risk (target)	1 = high-risk ward

- **Class balance.** 27.0% of wards are high-risk: 36.9% of rural wards against 18.9% of urban wards.
- **Data quality.** 64.4% of rural wards have at least one missing predictor, against 0.0% of urban wards. Symptom counts in about a third of rural wards are also under-reported (reduced by 45%), which is typical where clinics are few.
- **EDA findings.** High-risk wards have more elderly residents (14.1% against 12.1%), higher symptom rates (4.7 against 2.9), lower vaccination coverage (70.4% against

75.6%), fewer beds (1.38 against 1.87 per 1,000) and more past outbreaks (1.10 against 0.81). Travel inflow barely differs between the two groups.

- **Preparation steps.** Median imputation (fitted on training data only), standardisation for logistic regression, a stratified 70/30 split (360 test wards, 97 of them high-risk), and class weights to counter imbalance.

3. Models and Methodology

Four approaches were compared on the same split: a rule-based baseline that mimics current practice (flag a ward if its symptom rate exceeds 6), Logistic Regression, Random Forest and Gradient Boosting. Models were judged on precision, recall, F1-score, ROC-AUC and the number of false negatives, and then by recall within rural and urban wards. Because a missed high-risk ward is much more harmful, the decision threshold was tuned for recall instead of using the default 0.5.

4. Results and Evaluation

Model	Accuracy	Precision	Recall	F1	AUC	False neg.	Rural recall	Urban recall
Rule-based (symptom rate > 6)	73.9%	54.8%	17.5%	0.266	n/a	80	4.9%	38.9%
Logistic Regression	71.4%	48.1%	79.4%	0.599	0.814	20	82.0%	75.0%
Random Forest	74.7%	52.4%	68.0%	0.592	0.809	31	75.4%	55.6%
Gradient Boosting	79.4%	67.2%	46.4%	0.549	0.821	52	42.6%	52.8%

Table 2.1: Test-set performance at the default threshold of 0.5.

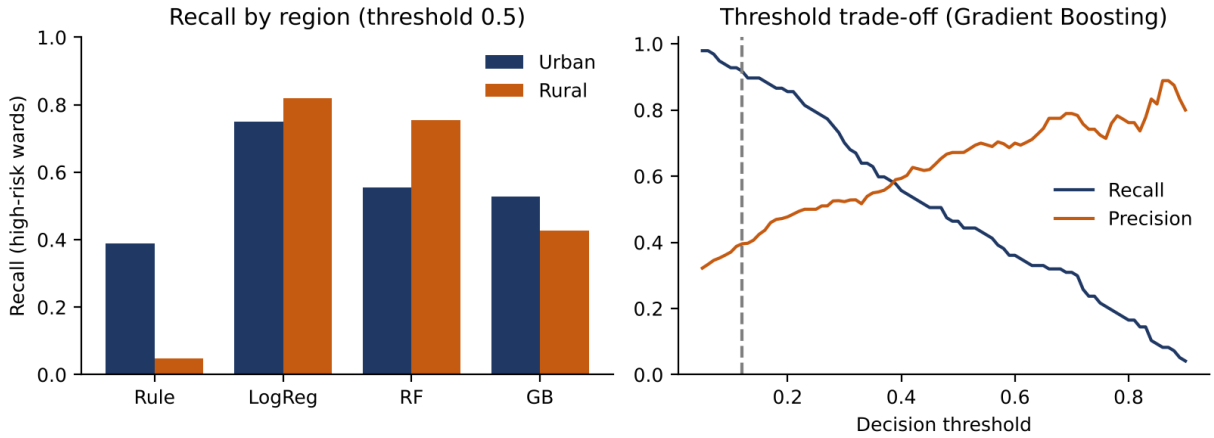


Figure 2.1: Left, recall by region at the default threshold. Right, recall and precision of Gradient Boosting as the decision threshold changes.

- **Baseline versus models.** The rule-based baseline finds only 17.5% of high-risk wards (80 of 97 missed), and only 4.9% in rural wards, because rural symptom counts are under-reported. All three models are much better; their AUC values are close (0.809 to 0.821), so model choice matters less than threshold choice.

- **Feature importance.** In the Random Forest, symptom rate (23.6%), beds per 1,000 (16.7%), elderly share (16.1%) and vaccination coverage (13.5%) are the main drivers, which agrees with the EDA and is easy to explain to officers.

Impact of false negatives

False negatives are the main risk: at the default threshold, Gradient Boosting misses 52 high-risk wards (53.6% of them) and Random Forest misses 31. If a missed ward is assumed to cost 8 times as much as an unnecessary alert, the total cost falls from 654 units for the rule-based baseline to 438 for Gradient Boosting at 0.5 and 200 after recall-oriented tuning.

Strategy	False neg.	False pos.	Cost units (FN = 8, FP = 1)	Recall
Rule-based baseline	80	14	654	17.5%
Logistic Regression at 0.5	20	83	243	79.4%
Random Forest at 0.5	31	60	308	68.0%
Gradient Boosting at 0.5	52	22	438	46.4%
Gradient Boosting, threshold 0.12	8	136	200	91.8%
Gradient Boosting, region-specific thresholds	8	133	197	91.8%

Fairness for rural and low-data wards

With one common threshold of 0.12, overall recall reaches 91.8%, but the burden is uneven: 82.9% of rural wards are flagged against 45.4% of urban wards, and recall is 96.7% for rural against 83.3% for urban wards. Rural wards are over-flagged and urban wards under-detected. Setting thresholds separately for each region (rural 0.21, urban 0.08) equalises detection: recall is 91.8% rural and 91.7% urban, with precision 49.6% rural and 30.3% urban. The mismatch in precision shows that rural wards give more reliable alerts here, so rural alerts should not be discounted as noise.

5. Decision and Resource-Allocation Policy

- **Chosen model.** Gradient Boosting, which has the highest AUC (0.821), with Logistic Regression run in parallel as a transparent cross-check (AUC 0.814). The model supports officers and does not decide alone.
- **Threshold policy.** Use region-specific thresholds chosen to give at least 90% recall in each region, because a missed high-risk ward costs far more than a false alarm.

The ranked wards are placed in three tiers, which drive resource allocation:

Tier	Risk score	Wards	High-risk wards inside	Allocation
Priority	≥ 0.35	111	61 (62.9% of all)	Pre-position ambulances, mobile medical teams and medicine stock; first call on staff
Watch	0.12 to 0.35	114	28 (28.9% of all)	Weekly surveillance, shared medicine stock, on-call staff, field verification
Routine	< 0.12	135	8 (8.2% of all)	Routine monitoring plus a random audit sample to catch misses

The Priority tier holds 30.8% of wards but 62.9% of the high-risk wards, so the scarcest resources (ambulances, specialist staff) go where they are most likely to be needed, while the Watch tier absorbs another 28.9%. Only 8.2% of high-risk wards fall into the Routine tier.

- **Protecting low-data wards.** Any ward with two or more missing indicators is placed in at least the Watch tier until a field survey completes its data; at least a fixed share of mobile-team visits is reserved for rural wards; and recall is reported separately for rural and urban wards each month.
- **Human oversight and monitoring.** District officers review every Priority-tier ward and can override the model with a recorded reason; the model is retrained after each outbreak season and checked for drift.
- **Limitations.** The data are simulated, the 8:1 cost ratio and the tier cut-offs are assumptions, and real labels may be affected by reporting bias, so the policy must be recalibrated on district data.

Question 4: Smart Waste Collection Prioritization System

1. Understanding of the Problem

A municipality collects waste on fixed schedules. Bins in crowded areas overflow before the truck arrives, while trucks waste fuel visiting bins that are nearly empty. The aim is to predict which bins will overflow soon and to turn those predictions into a daily collection route, without leaving remote or low-income areas under-served.

- **Machine-learning task.** Binary classification of each bin each day: will it overflow within the next 48 hours if it is not collected? The predicted probabilities then drive a route-prioritisation rule under a fixed vehicle capacity.
- **Key issues.** Overflow versus unnecessary trips; festival and rain effects on fill rate; noisy or missing sensor data in remote areas; and equitable service coverage.

2. Data Preparation and Exploratory Analysis

A simulated city of 240 smart bins on a 20 km by 20 km grid with the depot at the centre was created. 35.0% of bins are remote (more than 9 km from the depot) and 41.7% are in low-income areas. Daily fill rate averages 14.6% of capacity and rises by up to 60% in dense areas on festival days and falls by 12% on rainy days. One hundred and twenty days of history under the current fixed schedule gave 28,800 bin-day records, of which 14.0% were followed by overflow within 48 hours.

Feature	Description
sensor_fill	Latest sensor reading (% full); remote sensors are noisier and 30% of remote readings are stale
days_since	Days since the last collection (engineered)
density	Population density around the bin
waste_type	Domestic, market or mixed
festival, rain	Calendar and weather flags for the day
remote, low_income	Location and income-area flags (used for fairness checks)
dist_depot	Distance from the depot (km)
base_rate	Estimated daily fill rate from history

- **Preparation steps.** Days since collection and estimated fill rate were engineered; the data were split chronologically (first 75% to train, last 25% to test) to avoid leakage; features were scaled for logistic regression; class weights were used because only about one record in seven is a positive case.

3. Models and Methodology

Logistic Regression, Random Forest and Gradient Boosting were compared on precision, recall, F1 and AUC, and checked separately for remote and low-income bins. The best model

was then used in a 60-day simulation of three collection policies with the same fleet capacity of 60 bins per day (25.0% of bins):

- **Fixed schedule:** bins are visited in rotation, so every bin is collected every 4 days.
- **Data-driven (model only):** the highest-probability bins are collected each day, up to capacity, but only if the overflow probability is at least 0.35.
- **Data-driven with service guarantee:** as above, but remote or low-income bins that have waited 4 days or more are added first (up to 25% of daily capacity).

Routes are built with a nearest-neighbour heuristic from the depot. Fuel is estimated at 0.4 litres per km, which is an assumption.

4. Results and Evaluation

Model	Precision	Recall	F1	AUC
Logistic Regression	68.9%	95.6%	0.801	0.988
Random Forest	84.7%	97.3%	0.905	0.997
Gradient Boosting	94.1%	90.7%	0.924	0.996

Table 4.1: Overflow-prediction performance (chronological test set).

All three models work well because the sensor reading and days since collection are strong signals. Random Forest has the highest AUC (0.997) with 84.7% precision and 97.3% recall. Performance on remote bins is slightly lower (recall 95.1%, precision 81.4%) than on other bins (recall 97.9%, precision 85.8%), caused by the noisy remote sensors; low-income bins perform similarly to others (recall 97.7% against 96.9%).

Measure	Fixed schedule	Data-driven (model only)	Data-driven + service guarantee
Overflow rate (% of bin-days)	3.08%	0.23%	0.03%
Overflow bin-days	444	33	5
Missed-bin rate (% of bins that overflowed at least once)	20.0%	7.9%	2.1%
Distance travelled (km)	9,435	8,531	8,890
Fuel used (litres, at 0.4 L/km)	3,774	3,412	3,556
Bin visits	3,600	3,270	3,422
Unnecessary visits (bin under 30% full)	10.5%	2.1%	2.7%
Change in distance and fuel versus fixed	baseline	-9.6%	-5.8%

Table 4.2: Sixty-day simulation of the three policies (240 bins, 60 collections per day).

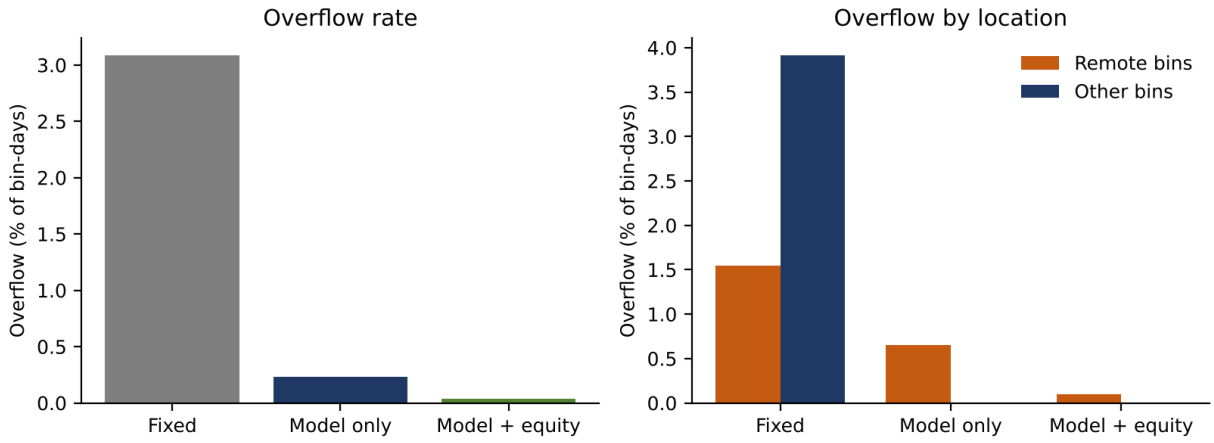


Figure 4.1: Left, overflow rate for each policy. Right, overflow rate in remote and other bins.

- **Data-driven versus fixed.** Moving from the fixed schedule to the data-driven policy cuts overflow from 3.08% to 0.23% of bin-days (a reduction of 93%), cuts unnecessary visits from 10.5% to 2.1%, and reduces distance and fuel by 9.6%. The same fleet, doing fewer but better-chosen visits, serves the city better.
- **Equity of service.** The model-only policy gives remote bins fewer collections (11.3 visits per bin in 60 days against 14.9 for other bins) and remote bins overflow 0.65% of bin-days against 0.00% elsewhere; low-income bins overflow 0.32% against 0.17% for the rest. This suggests that the noisy and stale remote sensors lower the model's confidence, so trucks are sent there less often. With the service guarantee, remote bins get 13.1 visits per bin and overflow falls to 0.10%; low-income bins overflow 0.02% of bin-days.
- **Cost of fairness.** The guarantee costs 4.2% more distance than the model-only policy (8,890 km against 8,531 km), but it still travels 5.8% less than the fixed schedule and cuts the missed-bin rate from 7.9% to 2.1%.

5. Decision and Recommendation

- **Recommended solution.** Deploy the Random Forest overflow model with the data-driven plus service-guarantee policy. It gives the lowest overflow (0.03%) and the lowest missed-bin rate (2.1%), uses 5.8% less fuel and distance than the fixed schedule, and keeps the service level of remote and low-income areas close to the rest of the city. It is both the most cost-effective and the most equitable of the three options.
- **Why not model-only?** It is the cheapest in distance, but it leaves remote bins with fewer visits and higher overflow, which is a fairness failure that a municipality cannot defend.
- **Operating rules.** Fall back to the fixed schedule for any bin whose sensor has been stale for more than two days; keep festival calendars as model inputs; review overflow and visit counts by ward every month; and prioritise sensor repair in remote areas, since sensor quality is the main weakness.
- **Limitations.** Costs and fuel use are assumptions, the city layout is simulated, and real routes follow roads rather than straight lines, so savings should be re-measured in a pilot before city-wide rollout.

Question 6: Smart Water Supply and Leakage Risk Assessment

1. Understanding of the Problem

The water board is losing more water every year, especially in older residential zones. It needs a model that flags pipeline segments likely to leak, so that limited inspection crews are sent to the right places first.

- **Machine-learning task.** Binary classification at pipeline-segment level (leak or no leak), with the leak probability used as a priority score.
- **Key issues.** An unnecessary inspection costs little but an undetected leak costs a great deal, so recall matters; the model must work in new areas as well as old; and new areas have newer sensors with data gaps and very few leaks to learn from.

2. Data Preparation and Exploratory Analysis

A simulated dataset of 3,000 pipeline segments was created, 1,659 (55.3%) in older zones and the rest in newly developed zones. Variables are pipe age, mean pressure, pressure variability, night flow (a classic leak indicator), consumption, repairs in the last five years, corrosive soil, rainfall and a zone flag.

- **Class balance.** 25.2% of segments are leak-prone: 42.8% in old zones and only 3.5% in new zones, so the new-zone sample has few positive cases.
- **Data quality.** 23.6% of new-zone segments have missing pressure or night-flow readings (recent sensor roll-out), against 0.0% in old zones.
- **EDA findings.** Leaking segments are much older (mean 40.0 years against 20.2), have more repairs (3.03 against 1.26), higher night flow (1.22 against 1.03) and more often lie in corrosive soil (39.4% against 26.9%). Mean pressure and consumption are almost identical for both groups.
- **Preparation steps.** Median imputation and scaling fitted on training data only, a stratified 70/30 split (900 test segments, 227 leaks, of which only 17 are in new zones), and class weights.

3. Models and Methodology

Logistic Regression, a depth-5 Decision Tree, Random Forest and Gradient Boosting were compared on Precision, Recall, F1-score and AUC, overall and separately for old and new zones. The decision threshold was then chosen to minimise total cost, using two assumed unit costs: Rs 5,000 per inspection and Rs 60,000 per undetected leak (lost water, damage and emergency repair).

4. Results and Evaluation

Model	Precision	Recall	F1	AUC	False pos.	False neg.	F1, old zones	F1, new zones
Logistic Regression	54.3%	83.3%	0.657	0.888	159	38	0.677	0.000
Decision Tree	48.7%	84.1%	0.617	0.845	201	36	0.642	0.133

Model	Precision	Recall	F1	AUC	False pos.	False neg.	F1, old zones	F1, new zones
Random Forest	60.5%	78.4%	0.683	0.881	116	49	0.706	0.000
Gradient Boosting	73.7%	55.5%	0.633	0.889	45	101	0.661	0.000

Table 6.1: Leakage-prediction performance at the default threshold of 0.5.

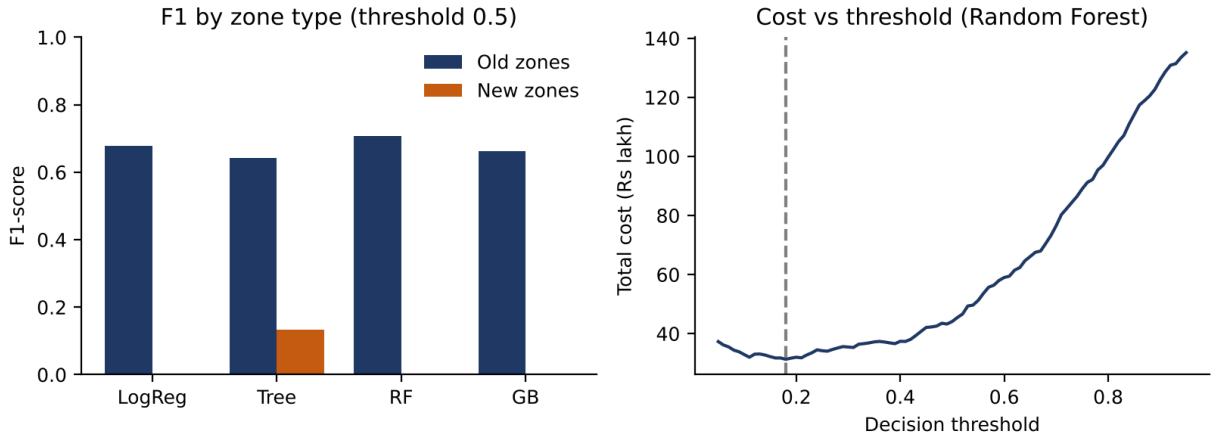


Figure 6.1: Left, F1-score in old and new zones. Right, total cost of inspections and undetected leaks as the threshold changes (Random Forest).

- **Model comparison.** Random Forest has the best F1 (0.683) with 60.5% precision and 78.4% recall. Gradient Boosting has the highest precision (73.7%) but misses 101 leaks, and Logistic Regression has the highest recall (83.3%) with more false alarms. The AUC values are all close (0.881 to 0.889 for the three best), so the choice depends on the cost of each error type.
- **Main drivers.** Pipe age (31.1% of Random Forest importance), the zone flag (14.7%), repair history (11.5%) and night flow (10.2%).

Cost of unnecessary inspections versus undetected leaks

A missed leak is assumed to cost 12 times as much as an inspection, so the default threshold of 0.5 is not cost-effective. The cost-optimal threshold is 0.18 for Random Forest. It raises recall to 95.2% (11 leaks missed) at the price of lower precision (43.7%, 278 unnecessary inspections).

Strategy (900 test segments)	Inspections	Leaks missed	Unnecessary inspections	Total cost
Inspect nothing	0	227	0	Rs 136.2 lakh
Inspect every segment	900	0	673	Rs 45.0 lakh
Inspect oldest 25% (age rule)	225	87	85	Rs 63.5 lakh
Random Forest at threshold 0.5	294	49	116	Rs 44.1 lakh
Random Forest at cost-optimal threshold 0.18	494	11	278	**Rs 31.3 lakh**

The cost-optimal model costs Rs 31.3 lakh, which is 30% less than inspecting every segment, 51% less than a simple age rule, and 29% less than the same model at 0.5.

Old zones versus new zones

At the default threshold, Logistic Regression, Random Forest and Gradient Boosting find none of the 17 leaks in new zones (F1 = 0), although they perform well in old zones (F1 0.71 for Random Forest). Even at the cost-optimal threshold, recall is 99.5% in old zones but only 41.2% in new zones (precision 24.1%). Choosing a separate cost-optimal threshold for each zone type (old 0.21, new 0.18) barely changes the result (cost Rs 31.2 lakh), which shows that the weakness comes from too few examples and missing sensor data in new zones, not from the threshold.

5. Decision and Maintenance-Priority Policy

- **Recommended model.** Random Forest with the cost-optimal threshold (0.18), because it gives the best F1 and the lowest total cost, and its importance scores are easy to explain to engineers.
- **Inspection priority.** Rank segments by leak probability and inspect in that order; where several have similar scores, favour segments with higher consequences (older pipes, repeated repairs, dense housing). If crews can inspect only a fixed share of the network, the ranking is what matters:

Share of segments inspected	Segments	Leaks found	Share of all leaks
Top 10%	90	71	31.3%
Top 20%	180	131	57.7%
Top 30%	270	170	74.9%

- **Replacement planning.** Pipes older than about 40 years with three or more repairs in five years go to a replacement shortlist, since repeated repairs on old pipes cost more than replacement.
- **New zones.** Do not rely on the model alone: complete the sensor roll-out (night-flow and pressure), raise alerts from simple night-flow rules, inspect a small random sample every year, and retrain a new-zone model once about 100 leak cases are available.
- **Governance.** Report recall and precision separately for old and new zones every quarter, recalibrate the unit costs with real figures, and keep a human engineer responsible for the final inspection list.

Question 8: Smart Disaster Early-Warning Reliability System

1. Understanding of the Problem

A coastal district receives rainfall, river, dam, soil, cyclone and satellite information, and must decide early which zones to warn and evacuate. This solution focuses on flood risk at zone level; the same pipeline applies to cyclone and landslide risk.

- **Machine-learning task.** Binary classification of each zone (will it flood in the forecast window?), evaluated at warning lead times of 6, 12, 24 and 48 hours, plus a priority score for evacuation support.
- **Key issues.** A missed flood is far worse than a false alarm; forecasts become less accurate the earlier the warning is issued, so there is a trade-off between lead time and reliability; and low-lying, dense, low-income zones with little evacuation capacity must not be treated as just another zone.

2. Data Preparation and Exploratory Analysis

A simulated dataset of 4,000 zone records was created with 24-hour rainfall, river level, dam discharge, soil moisture, cyclone alert level (0 to 3), elevation, population density and evacuation-centre capacity ratio (capacity divided by people at risk). 30.9% of zones are classed as vulnerable (lower elevation, higher density, less evacuation capacity). 12.1% of records are flood events: 16.2% in vulnerable zones against 10.2% elsewhere.

- **EDA findings.** Flooded zones show higher rainfall (84.7 mm against 51.7 mm), higher river level (3.96 m against 3.09 m), higher dam discharge (1,098 against 894), stronger cyclone alerts (1.33 against 0.58) and lower elevation (7.9 m against 13.1 m).
- **Lead-time simulation.** To measure warning lead time honestly, the model inputs were degraded to mimic forecasts issued 6, 12, 24 and 48 hours ahead: the noise added to rainfall, river level, dam discharge, soil moisture and cyclone level grows with the square root of the lead time.
- **Preparation.** Stratified 70/30 split (1,200 test records, 145 flood events), standardisation for logistic regression and class weights.

3. Models and Methodology

Three machine-learning models (Logistic Regression, Random Forest, Gradient Boosting) were compared with a rule-based baseline that raises an alarm when the river level exceeds 4.0 m. Evaluation used recall (the share of real events detected, equal to 1 minus the missed-event rate), precision, F1 and AUC, followed by tuning of the alert threshold for the warning lead time and by a vulnerability check.

4. Results and Evaluation

Model	Precision	Recall	F1	AUC	Missed-event rate	Recall, vulnerable	Recall, others
River-level rule (> 4.0 m)	22.7%	51.7%	0.315	n/a	48.3%	50.8%	52.4%

Model	Precision	Recall	F1	AUC	Missed-event rate	Recall, vulnerable	Recall, others
Logistic Regression	38.3%	76.6%	0.510	0.889	23.4%	84.1%	70.7%
Random Forest	49.1%	58.6%	0.535	0.862	41.4%	60.3%	57.3%
Gradient Boosting	62.9%	38.6%	0.479	0.867	61.4%	38.1%	39.0%

Table 8.1: Performance at 24 hours lead time and the default threshold of 0.5.

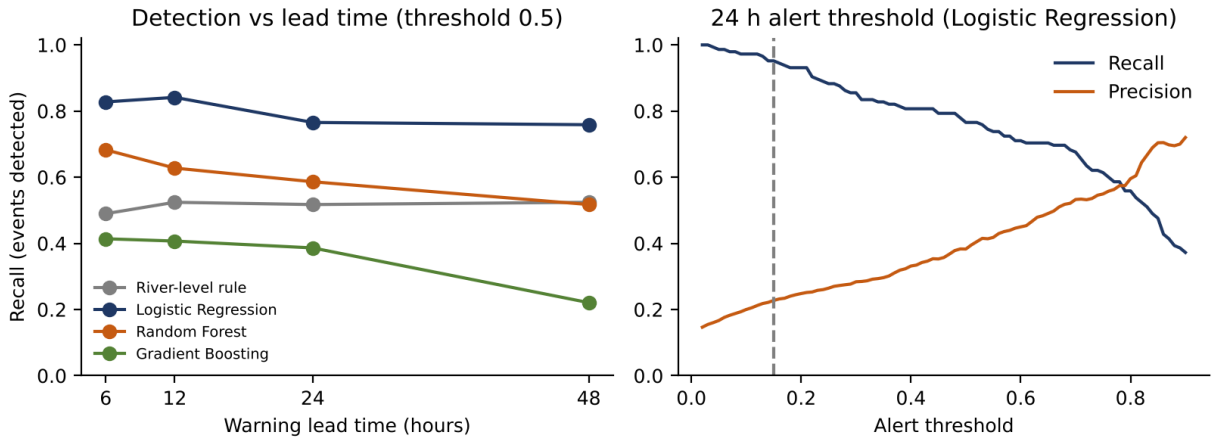


Figure 8.1: Left, event detection (recall) at the default threshold for each lead time. Right, recall and precision of the chosen model at 24 hours as the alert threshold changes.

- **Baseline versus models.** The river-level rule misses 48.3% of flood events at 24 hours because rainfall, dam discharge, cyclone level and elevation all matter. Logistic Regression detects 76.6% at the default threshold (missed-event rate 23.4%) and has the best AUC (0.889). Gradient Boosting has high precision (62.9%) but misses 61.4% of events at the default threshold, which is unacceptable for early warning.

Warning lead time and missed-event rate

The detection ability of the machine-learning models falls as the warning is issued earlier, because the forecast inputs are less reliable. Table 8.2 shows the chosen model with its threshold re-tuned at each lead time to keep at least 95% recall.

Lead time	AUC (threshold 0.5)	Chosen threshold	Missed-event rate	Precision	Share of alerts that are false
6 hours	0.933	0.19	4.1%	29.8%	70.2%
12 hours	0.919	0.21	4.1%	27.8%	72.2%
24 hours	0.889	0.15	4.8%	22.7%	77.3%
48 hours	0.865	0.19	4.8%	21.6%	78.4%

To keep the missed-event rate near 5%, the share of false alerts rises from 70.2% at 6 hours to 78.4% at 48 hours. Longer lead time is therefore bought with more false alarms.

Do vulnerable communities receive adequate priority?

At the chosen 24-hour threshold of 0.15, recall is 98.4% in vulnerable zones and 92.7% in other zones, so the model does not neglect vulnerable communities in detection. However, the ordering of the alerted zones matters for evacuation. If zones are ranked by flood probability alone, only 53.3% of the top 60 are vulnerable zones. A priority score that multiplies flood probability by a vulnerability factor, the evacuation-capacity gap and the population exposed puts 100.0% vulnerable zones in the top 60; these zones have a mean evacuation-capacity ratio of 0.41 against 0.75 for all alerted zones, and 65.0% of them did flood.

5. Decision: Alert Threshold and Evacuation-Support Plan

- **Chosen model and threshold.** Logistic Regression with an alert threshold of 0.15 at 24 hours. This keeps the missed-event rate at 4.8% (7 of 145 events missed) while precision is 22.7%; 50.6% of zones are alerted. The threshold is deliberately low because a missed flood can cost lives, whereas a false alarm costs only preparation effort.

To make this many alerts workable, alerts are issued in three tiers instead of one:

Tier	Rule (24 h model)	Zones	Floods inside	Action
Evacuate	$p \geq 0.50$	290	111 (38.3% of zones)	Start evacuation of the highest-priority vulnerable zones; open shelters
Prepare	$0.15 \leq p < 0.50$	317	27 (8.5% of zones)	Pre-position boats, pumps and staff; alert community volunteers; re-check every 6 hours
Monitor	$p < 0.15$	593	7 (1.2% of zones)	Routine monitoring

- **Staged lead times.** Issue a Prepare-level warning at 48 hours, a formal warning at 24 hours and an evacuation order at 12 hours or less, where the forecast is more reliable and false alarms are fewer.
- **Evacuation priority.** Within the alerted zones, rank by flood probability × vulnerability × evacuation-capacity gap × people exposed, and allocate buses, shelters and rescue teams in that order, with a reserved share for zones that have the lowest capacity.
- **Human oversight.** The district disaster-management officer confirms every Evacuate-tier order, using local knowledge and satellite images, and can lower a tier only with a recorded reason.
- **Limitations.** The data are simulated and flood events were predicted from the same measured variables that generated them, so real performance will be lower; thresholds must be re-validated against past floods, and the model should be retrained after every monsoon.

Question 10: Smart Energy Demand Forecasting for Public Buildings

1. Understanding of the Problem

Government offices, schools, hospitals and transport depots use electricity that depends on occupancy, weather, working hours and equipment, and energy is wasted because consumption is not actively monitored. The task is to forecast hourly electricity demand, estimate the savings of energy-saving actions, and judge whether solar generation and batteries improve cost and reliability.

- **Machine-learning task.** Regression: forecast each building's hourly demand (kW) for the next day.
- **Key issues.** Different buildings have very different load shapes; demand peaks drive both tariffs and grid stress; weather and occupancy matter; and hospitals need reliable supply.

2. Data Preparation and Exploratory Analysis

A simulated hourly dataset covering one year (2025) for four public buildings was created: a government office, a school, a hospital and a transport depot. After building lag features, 34,368 hourly records remained. Mean demand is 30.5 kW for the office, 13.1 kW for the school, 188.2 kW for the hospital and 67.5 kW for the depot; hospital peak demand reaches 426 kW.

Feature	Description
hour (sine/cosine), day of week, month, holiday	Calendar effects; hour is encoded cyclically so 23:00 is close to 00:00
temp, humidity	Weather; temperature drives cooling load
occupancy	Share of normal occupancy (0 to 1.3)
building	Building identifier
lag24, lag168	Demand 24 hours and 168 hours (one week) earlier
solar	Normalised solar generation profile, used for the solar analysis

- **EDA findings.** The government office uses 34.2 kW on weekdays and 21.3 kW at weekends, while the hospital and depot run almost continuously. Demand is positively correlated with temperature (correlation 0.32 across all buildings), and demand follows a clear working-hours pattern with a distinct daily peak.
- **Preparation steps.** Missing lag values from the first week were dropped; the split is chronological to avoid leakage: January to October for training (28,512 records) and November to December for testing (5,856 records). In deployment, forecast weather would replace the measured temperature used here.

3. Models and Methodology

Linear Regression, Random Forest and Gradient Boosting were compared with a seasonal-naive baseline that repeats last week's demand for the same hour. Metrics were MAE, RMSE and MAPE, plus two peak measures: the share of days on which the forecast peak hour is within one hour of the real peak hour, and the average error in peak size. Savings from actions were then estimated on the test period with an assumed time-of-day tariff of Rs 5/kWh at night (22:00 to 06:00), Rs 7/kWh in the day and Rs 9.5/kWh at the evening peak (18:00 to 22:00).

4. Results and Evaluation

Model	MAE (kW)	RMSE (kW)	MAPE	Peak-hour hit rate	Peak size error	MAE in top-10% hours (kW)
Seasonal naive (same hour last week)	6.07	13.40	11.1%	50.4%	11.1%	13.58
Linear Regression	16.78	19.24	100.3%	62.7%	33.9%	20.58
Random Forest	1.59	2.39	4.5%	80.3%	3.3%	2.06
Gradient Boosting	1.80	2.78	4.9%	78.7%	4.5%	3.14

Table 10.1: Forecast accuracy on the November–December test period (all buildings).

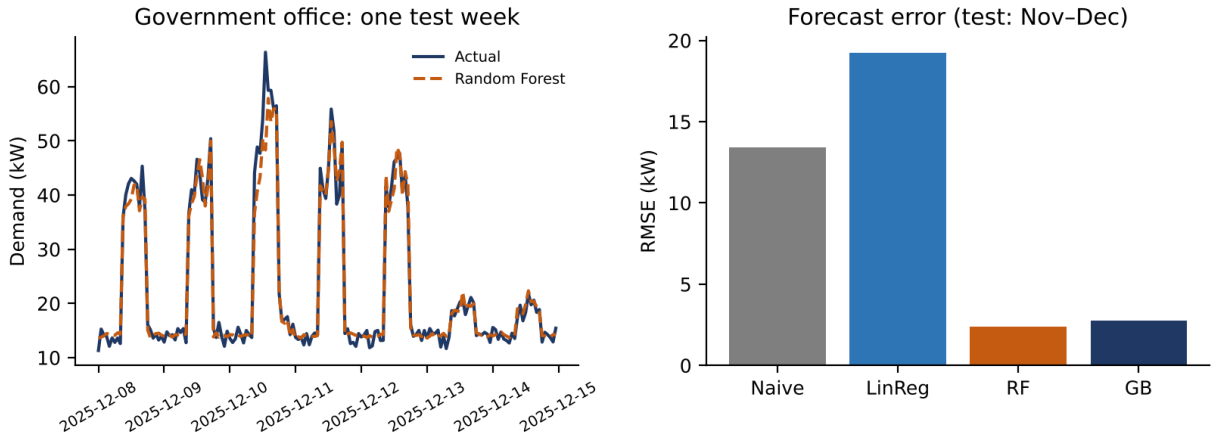


Figure 10.1: Left, actual and forecast demand for one week for the government office. Right, RMSE of each model.

- **Model comparison.** Random Forest is the most accurate: MAE 1.59 kW, RMSE 2.39 kW and MAPE 4.5%, compared with RMSE 13.40 kW for the seasonal-naive baseline (82% lower) and 19.24 kW for Linear Regression. Linear Regression performs poorly (MAPE 100.3%) because a single linear model cannot represent four buildings with different, non-linear load shapes. Gradient Boosting is close behind (RMSE 2.78 kW).
- **Peak-demand accuracy.** Random Forest finds the hour of the daily peak within one hour on 80.3% of building-days (baseline 50.4%) and its peak size is off by only 3.3% (baseline 11.1%), and its error in the top 10% of hours is 2.06 kW, which matters most for peak charges.

- **By building.** Random Forest MAPE is 6.4% for the office, 6.5% for the school, 2.3% for the hospital and 2.7% for the depot, so accuracy is acceptable everywhere. The most important inputs are the previous day's and previous week's load (71.3% and 16.1%) and temperature (11.8%).

Electricity-cost savings from each proposed action

The test-period cost without any action is Rs 22.6 lakh for 330,295 kWh. The table shows the simulated saving of each action under the assumptions listed below it.

Action	Energy saved	Cost saved (Rs)	Cost saved (% of baseline)
A. HVAC set-point +1 °C (forecast-guided)	1.0%	25,475	1.13%
B. Off-hours shutdown of idle equipment	0.9%	19,162	0.85%
C. LED + occupancy controls	7.4%	168,127	7.44%
D. Forecast-driven peak shifting (18–22 h)	0.0%	6,655	0.29%
A+B+C combined	9.3%	212,764	**9.41%**

Assumptions: A reduces weather-driven cooling load by 12% (about 1 °C higher set-point, guided by the day-ahead forecast); B cuts 12% of idle-hour load in non-hospital buildings; C cuts 8% of load during occupied hours; D shifts 5% of evening-peak load to the afternoon. The test period is November to December, so cooling savings are understated compared with summer.

LED and occupancy controls (C) give the largest saving, 7.44% of cost, followed by set-point control (1.13%) and off-hours shutdown (0.85%). Peak shifting (D) saves only 0.29% of cost because the tariff difference between the afternoon and the evening peak is small, but it lowers demand during the grid's busiest hours. The three efficiency actions together save 9.41% (Rs 212,764).

Do solar energy and batteries improve cost and reliability?

Solar and battery sizes were set relative to each building's average demand, applied after the efficiency actions: PV of roughly 0.7 to 1.2 kW peak per kW of mean load, and a battery holding four hours of average load. The battery charges from surplus solar and cheap night power and discharges in the 18:00 to 22:00 peak. Reliability was tested with twelve random six-hour grid outages, where the hospital must be supplied in full and the other buildings must supply 30% of their load (critical services only).

Building	PV only: cost saving	PV + battery: cost saving	Peak demand (kW): before	Peak grid import with PV + battery (kW)	Outage load served by PV + battery
Government office	22.1%	28.5%	69.2	61.4	99.1%
School	28.0%	33.8%	36.7	29.8	99.8%
Hospital	17.3%	23.9%	248.3	204.9	72.3%
Transport depot	24.5%	31.1%	81.2	79.0	100.0%

Across the four buildings, solar alone cuts the remaining electricity cost by 20.1%, and solar with a battery by 26.7%. Solar plus battery also keeps critical loads running through outages: the office, school and depot are fully supplied, while the hospital, which must supply its full load, is covered for 72.3% of outage demand, so it still needs a diesel generator or a larger battery.

5. Final Energy-Management Recommendation

- **Forecasting model.** Deploy the Random Forest day-ahead forecast (MAPE 4.5%) in a monitoring dashboard that compares actual with forecast demand every hour and flags deviations above a set margin as possible wastage.
- **Actions, in order.** (1) LED and occupancy controls: the largest and most certain saving; (2) forecast-guided cooling set-points and off-hours shutdown, especially in offices and schools; (3) solar with a battery, starting with the hospital and transport depot where load is high and continuous; (4) peak shifting to reduce stress on the grid and any future demand charges.
- **Total effect.** The efficiency actions save about 9.4% of electricity cost on these test months, and solar with a battery removes a further 26.7% of what remains, with better resilience during outages.
- **Limitations.** The demand, weather and tariff are simulated; the saving percentages for each action and the PV and battery sizes are assumptions; and the cost of the equipment itself was not modelled, so a payback analysis is needed before purchase.